

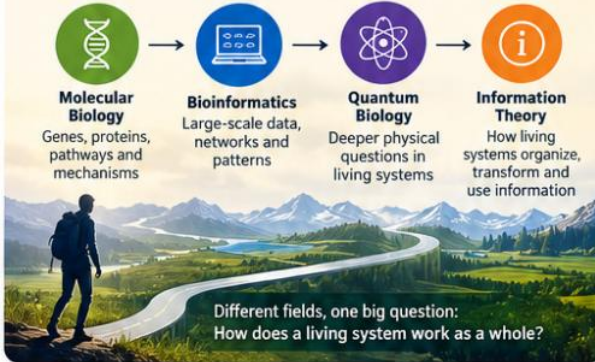
WHY I THINK BIOLOGY NEEDS A NEW THEORY

From Molecular Biology to Bioinformatics,
Quantum Biology and Information

LIFE IS MORE
THAN A LIST
OF MOLECULES —
IT IS AN ORGANIZED,
DYNAMIC SYSTEM
THAT TRANSFORMS
INFORMATION.

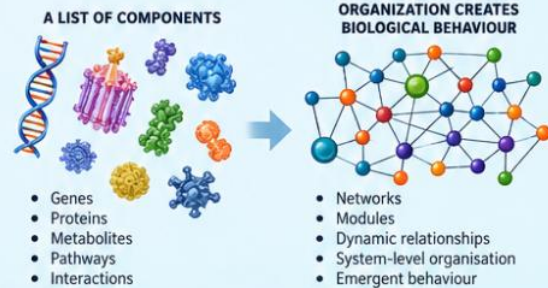
1 A SCIENTIFIC JOURNEY

My perspective on biology has evolved through different fields, each opening new questions.



2 FROM COMPONENTS TO ORGANIZATION

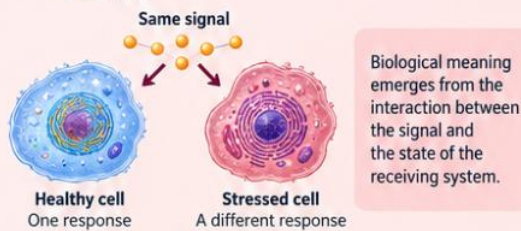
Knowing all the parts is not the same as understanding the whole. Relationships and organization matter.



The same components can behave very differently depending on how they are organized.

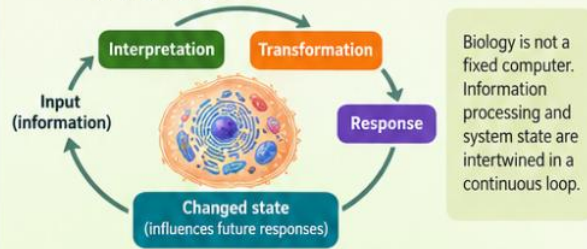
3 CONTEXT CHANGES MEANING

The same signal can produce different responses depending on the state, history and environment of the cell.



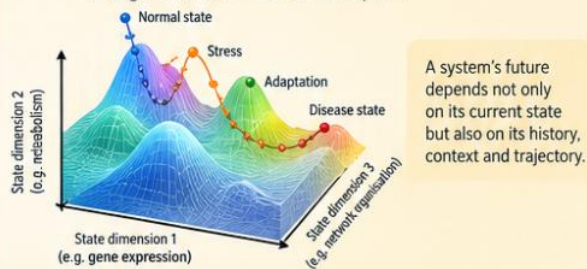
4 BIOLOGY IS DYNAMIC

A living system continuously processes information while changing its own state.



5 BIOLOGICAL STATES AND TRAJECTORIES

A biological system can be represented as a trajectory through a multidimensional state space.



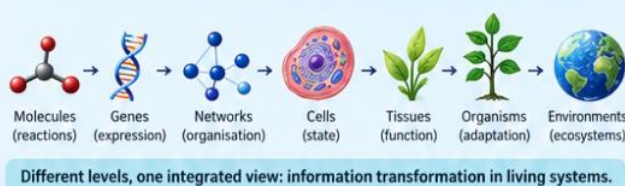
6 TOWARDS NEW SYSTEM-LEVEL OBSERVABLES

Beyond individual genes, we can measure properties of the whole system.



7 CONNECTING SCALES

A new theory should link information across biological levels.



8 WHY THIS MATTERS

A broader information framework could help us:

- Understand stress and adaptation in plants
 - Detect early signs of disease
 - Explore when quantum effects are biologically relevant
 - Develop predictive and testable models
 - Support applications in agriculture, medicine and beyond
- The goal is not to replace molecular biology, but to build on it with a deeper understanding of how living systems organize, transform and use information.



THE BIG IDEA

Perhaps biology needs a new theory —
not just of information, but of information transformation
in self-organizing, self-changing living systems.

Science advances not only by discovering more things, but also by finding better ways to describe how life remains organized while constantly changing.

There are moments in a scientific career when the questions begin to change.

At the beginning, the questions are usually quite concrete.

What is this gene doing?

Which protein is involved?

Why does this pathway change under stress?

Which genes are expressed differently?

Can we identify the molecular mechanism?

These are good questions. In fact, they are the questions on which modern biology has been built.

I have spent a significant part of my scientific life asking questions like these.

My training was in molecular biology. I worked with genes, proteins, plants and biological responses. Later, bioinformatics changed the way I looked at biological systems. Instead of studying one molecule or one pathway at a time, I could begin looking at thousands of genes simultaneously, and eventually at networks and patterns.

Then came quantum biology.

Then information theory.

And gradually, almost without my noticing it, the nature of the question changed.

I was no longer asking only:

“What molecule is doing what?”

I was increasingly asking:

“How does a living system organize, transform and use information?”

That question has taken me to some rather unexpected places.

It has made me think about cells as information-processing systems, about intelligence outside the brain, about collective behaviour, about quantum phenomena in biology and, eventually, about consciousness.

But this article is not about consciousness.

Nor is it an attempt to repeat the arguments of the previous nine articles.

Those articles were a journey.

This article is about what that journey has made me think.

And what I have come to suspect is that biology may eventually need something more than another collection of discoveries.

It may need a **new theoretical framework**.

I do not mean that molecular biology is wrong.

I do not mean that chemistry is inadequate.

I do not mean that everything we know about genes, proteins, pathways and biochemical reactions needs to be discarded.

Quite the opposite.

The molecular foundations of biology are extraordinarily successful.

But perhaps they are not the whole story.

Perhaps we have reached a stage in biology similar to what happens in many sciences: after learning an enormous amount about the components, we begin to ask whether we also need a better language for describing the **relationships among the components**.

That is the question behind this article.

And it is also, in many ways, the question behind my own scientific journey.

A question that stayed with me

When I first entered molecular biology, I did not think I would one day be writing about information theory or quantum biology.

Science does not usually happen that way.

You begin with a problem.

Then another problem appears.

Then a technique gives you a different perspective.

Then a new field opens a door.

Eventually, you discover that the questions you are asking are no longer quite the questions with which you started.

Molecular biology taught me to think in terms of molecules.

Genes.

Proteins.

RNA.

Enzymes.

Metabolites.

Interactions.

Pathways.

The logic was wonderfully tangible.

A gene is expressed.

RNA is produced.

A protein is synthesized.

The protein interacts with another molecule.

A pathway is activated.

A phenotype emerges.

This molecular language transformed biology.

But as biological datasets became larger, something interesting happened.

The more data we generated, the harder it sometimes became to tell the biological story.

We could identify thousands of differentially expressed genes.

We could construct enormous interaction networks.

We could identify enriched pathways.

We could build increasingly sophisticated models.

Yet a question remained:

What is the state of the biological system as a whole?

That question is not always answered by knowing the state of each component separately.

And perhaps this is where my journey from molecular biology to bioinformatics became important.

From molecules to thousands of measurements

Bioinformatics changes your relationship with biological information.

A molecular biologist can become deeply familiar with a particular gene, protein or pathway.

A bioinformatician is often confronted with something very different.

Thousands of genes.

Thousands of samples.

Millions of variants.

Complex interaction networks.

Multiple dimensions.

Noise.

Missing data.

Batch effects.

Biological heterogeneity.

And, increasingly, time.

Suddenly the biological system looks less like a list and more like a landscape.

You begin to see patterns that are difficult to describe in the language of individual molecules.

A gene rarely acts alone.

A protein rarely exists in isolation.

A pathway rarely operates independently of other pathways.

And a phenotype rarely has a single cause.

Biology is interconnected.

We have known this for a long time.

But modern data have made the interconnectedness impossible to ignore.

This is one of the reasons systems biology emerged as such an important way of thinking.

The question shifts from:

“What does this component do?”

to:

“How does this component participate in the behaviour of the system?”

That sounds like a small change.

It is not.

It changes what we consider to be an explanation.

The problem with knowing all the parts

Imagine that we somehow had a perfect catalogue of every molecule inside a cell.

Every gene.

Every transcript.

Every protein.

Every metabolite.

Every modification.

Every interaction.

Would we then completely understand the cell?

My instinct is that the answer is no.

And this is not because the molecules are unimportant.

It is because **relationships matter**.

Consider something as simple as a network.

Suppose I give you a list of 100 people.

You know their names.

You know their ages.

You know their professions.

You know where they live.

You know almost everything about them individually.

But I do not tell you who talks to whom.

You still do not know much about the social organization of the group.

Now I give you the connections.

Suddenly the picture changes.

You can see communities.

You can see central individuals.

You can see isolated individuals.

You can see clusters.

You can see bridges between groups.

You can see information pathways.

The individuals have not changed.

The organization has become visible.

Biology may be similar.

The molecules are the components.

The organization is the pattern of relationships among them.

And the biological behaviour may depend on both.

This is one reason I increasingly find myself uncomfortable with descriptions of biological systems that stop at lists of components.

A list is not an organization.

A network is not merely a list of nodes.

And a living system is not merely a list of molecules.

Biology has another complication: context

There is another reason why biological information is unusual.

The meaning of a biological signal depends heavily on context.

A signal does not arrive in an empty system.

It arrives in a cell that already has a history.

It has already experienced other signals.

Its metabolic state is different from that of another cell.

Its location may be different.

Its neighbours may be different.

Its internal network may be configured differently.

The same external stimulus can therefore have very different consequences.

This idea was already central to the argument developed at the beginning of the *Rethinking Biology* series: conventional information theory gives us powerful tools for quantifying uncertainty, communication and transmission, but biological systems introduce additional questions concerning context, state, history, spatial organization and interpretation.

The point is not that Shannon's information theory is inadequate.

That would be an unfair interpretation.

Shannon deliberately abstracted away from meaning because that abstraction was precisely what made his mathematical framework so powerful.

The question is different:

What happens when we bring information theory into a system in which the response to information depends on the state of the receiver?

That is where biology becomes interesting.

The same message can mean different things

Let us imagine two cells.

Cell A is healthy.

Cell B is under severe stress.

Now imagine that both receive exactly the same molecular signal.

If we think only about transmission, we might say:

The signal was the same.

The information transmitted was the same.

But the biological consequences may be completely different.

Why?

Because the receiving systems are different.

Their internal states are different.

Their histories are different.

Their networks may be configured differently.

Their metabolic conditions may be different.

Their previous exposure to signals may be different.

Their spatial environments may be different.

So perhaps the biological meaning of a signal is not contained entirely in the signal itself.

Perhaps meaning emerges from an **interaction between signal and state**.

This is a deceptively simple idea.

But if taken seriously, it has major consequences.

It means that understanding biological information may require us to understand not only what is transmitted but also **what receives it, how the receiver is organized, what state it is in, and what happened before the signal arrived**.

That begins to look less like conventional communication theory and more like a theory of **information transformation in living systems**.

From information transmission to information transformation

This distinction has become increasingly important to me.

We often speak about information as if it were something that travels.

A receptor receives a signal.

A signalling pathway transmits it.

A transcription factor responds.

A gene is expressed.

But the cell is not a passive communication channel.

The cell changes the signal.

It interprets it.

It combines it with other signals.

It modifies its response according to its internal state.

And the response itself changes the state of the cell.

So we have something more complicated than:

Input → Output

We have something closer to:

Input → interpretation → transformation → response → changed state → new interpretation

And then the cycle begins again.

The system is changing while it is processing information.

This is one of the central ideas I would like to explore in a future theory of biological information.

Perhaps biology is not fundamentally about information moving through a fixed machine.

Perhaps it is about information being transformed by a machine that is **continuously rebuilding and reorganizing itself**.

That is a very different problem.

A living system is not a fixed computer

This is also where comparisons between biology and artificial intelligence need to be made carefully.

An artificial neural network normally has a defined architecture.

Its parameters may change during training.

But the physical system executing the model does not ordinarily rebuild itself because of every incoming signal.

Biological systems are different.

A cell can change its molecular composition.

It can alter its membrane.

It can change gene expression.

It can modify proteins.

It can alter metabolic pathways.

It can change connections.

It can grow.

It can divide.

It can die.

An organism can learn from experience and modify its future responses.

Evolution operates over still longer timescales.

So biological information processing occurs in a system whose **hardware and software are intertwined**.

The distinction between hardware and software becomes blurry.

The system processes information, but the processing can change the system itself.

And then the changed system processes subsequent information differently.

This creates a loop:

Information changes state. State changes information processing. Information processing changes future state.

Perhaps this loop is one of the most fundamental characteristics of living systems.

What, then, is a biological state?

This question has increasingly fascinated me.

We routinely talk about biological states.

A cell is in a proliferative state.

A cell is differentiated.

A cell is stressed.

A cell is senescent.

A tissue is diseased.

An organism is adapting.

But these descriptions are often qualitative.

We know what we mean biologically, but can we define these states mathematically?

Can we represent them in a quantitative state space?

Imagine a multidimensional space in which every point represents a possible biological configuration.

One axis might represent gene-expression organization.

Another might represent metabolic state.

Another network connectivity.

Another variability.

Another spatial organization.

Another temporal dynamics.

A cell would then not simply be “healthy” or “stressed.”

It would occupy a particular region of this multidimensional space.

As conditions change, the cell would move through that space.

Stress might correspond not to one gene changing, but to a trajectory through biological state space.

Recovery might correspond to a movement back toward another region.

Disease might involve a transition into a different attractor-like region.

Differentiation might represent a relatively stable transition between states.

This is not a complete theory.

It is a way of asking a question.

Can biological states be represented as measurable positions and trajectories in an information-organizational space?

If the answer is yes, we might gain a new way of studying biology.

The importance of trajectories

Biology is dynamic.

Yet much biological analysis is still based on snapshots.

We take a sample.

We sequence it.

We compare it with another sample.

We identify differences.

This is enormously valuable.

But a snapshot tells us where the system is.

It does not necessarily tell us where it is going.

Consider two cells that have identical current expression profiles.

They may nevertheless have different histories.

One cell may have arrived at its current state gradually.

The other may have arrived there following a severe perturbation.

Their future trajectories might therefore be different.

In mathematical terms, the current state alone may not completely determine the future behaviour if relevant historical information is not captured in the state representation.

This brings us back to a concept that appears repeatedly in thinking about biological information:

history matters.

The biological system carries traces of its past.

Some of these traces are molecular.

Some are epigenetic.

Some are structural.

Some are network-level.

Some may involve long-lived cellular states.

The precise mechanisms differ.

But the broader idea is powerful:

A living system has a memory of what has happened to it.

A theory of biological information that ignores history may therefore miss an important part of biology.

Perhaps biology needs new observables

This is where my own research thinking begins to move from philosophy toward measurement.

If biological organization matters, then we should be able to measure something about it.

That sounds obvious.

But it is actually a demanding requirement.

It is easy to say:

“Organization matters.”

It is much harder to answer:

“What is organization?”

“How do we quantify it?”

“What units does it have?”

“How robust is the measurement?”

“What does it predict?”

“Does it generalize across datasets?”

“Does it distinguish biological states?”

“Does changing it experimentally alter biological function?”

These are the questions that transform an interesting idea into a research programme.

And this is why I am increasingly interested in **system-level observables**.

Instead of looking only at individual genes, perhaps we should also measure properties of the whole expression system.

Instead of asking only which genes change, perhaps we should ask how the **relationships among genes change**.

Instead of measuring only abundance, perhaps we should measure organization.

Instead of looking only at static expression, perhaps we should look at trajectories.

Instead of asking only which pathway is enriched, perhaps we should ask whether the architecture of information processing has changed.

A different way of looking at transcriptomic data

This way of thinking becomes particularly interesting with transcriptomics.

Modern RNA-seq datasets are often extraordinarily rich.

We can compare thousands of genes across multiple conditions.

We can construct correlation matrices.

We can infer networks.

We can perform dimensionality reduction.

We can calculate entropy.

We can examine variability.

We can identify modules.

But perhaps these tools can be brought together around a deeper question:

What is the organizational state of the transcriptome?

This is subtly different from asking which genes are expressed.

Imagine two transcriptomes with similar average expression levels.

They could nevertheless have very different internal organization.

In one, genes may fluctuate together in highly coordinated patterns.

In another, expression may be much more independent.

One may have a strongly structured network.

The other may be comparatively fragmented.

One may have relatively low variability.

The other may be highly variable.

If these differences correlate with biological function, then the organizational properties of the transcriptome themselves become meaningful biological observables.

This is an area in which information theory and network analysis can potentially contribute something beyond conventional differential expression.

From gene lists to relationships

There is a psychological trap in high-throughput biology.

When we have a table containing thousands of genes, it is natural to focus on the individual entries.

Gene 1.

Gene 2.

Gene 3.

Gene 4.

And so on.

But perhaps the more interesting information is sometimes contained between the entries.

Gene A and Gene B are correlated.

Gene C and Gene D form a module.

Gene E changes only when Gene F is active.

A group of genes becomes coordinated under one condition but not another.

These relationships may reveal something about the state of the system that individual measurements do not.

This is why network thinking is so powerful.

But I would like to take the idea one step further.

Networks themselves have states.

A network can become more integrated.

More fragmented.

More centralized.

More modular.

More variable.

More coherent.

More disordered.

These properties may be measurable.

And if they are biologically meaningful, they could become part of a theory of biological information.

Entropy is not the enemy of organization

Another concept that becomes important here is entropy.

Entropy has a long and complicated history across physics, information theory and biology.

In information theory, entropy provides a measure of uncertainty or variability in a probability distribution.

In biological systems, entropy-based measures can sometimes provide useful information about diversity, disorder or uncertainty.

But an important question arises:

Does more entropy always mean less biological organization?

Not necessarily.

A living system is not simply trying to minimize disorder.

Biological systems operate between stability and variability.

Too little variability may make a system rigid.

Too much variability may make it unstable.

Adaptation may require a controlled degree of flexibility.

A useful biological theory therefore may need to describe both **organization and variability**.

Perhaps the interesting region is not maximum order or maximum disorder.

Perhaps it lies somewhere between them.

This is one reason why I find it useful to think about biological systems in terms of multiple complementary dimensions rather than searching for a single measure of “order.”

Why correlation alone may not be enough

There is another problem that becomes apparent once we begin thinking this way.

A correlation matrix is useful.

But correlation does not automatically tell us biological meaning.

Two genes may correlate because they are controlled by a common factor.

A third variable may influence both.

The relationship may disappear under another condition.

Or it may be a consequence rather than a cause.

This means that any proposed organizational measure needs careful validation.

A theory of biological information should therefore not stop at descriptive statistics.

It should make predictions.

Suppose we identify a particular organizational property that consistently changes during stress.

The next question should be:

Does manipulating that organization change the biological outcome?

That is where perturbation becomes important.

From observing systems to perturbing systems

There is a fundamental difference between watching a system and disturbing it.

Suppose I observe that two genes are strongly connected during stress.

I can report that association.

But I learn much more if I perturb one component and observe what happens to the organization of the system.

Does the network reorganize?

Does the system move to a new state?

Does the phenotype change?

Does the predicted information flow change?

Does the system compensate?

Does another pathway take over?

This is where a theory becomes much more interesting.

A good theory should tell us:

If I change this, what should happen next?

That is a much stronger scientific test than simply showing that two variables are associated.

Biological organization may be causal

This leads to a difficult philosophical question.

Is organization merely a consequence of molecular interactions?

Or can organization itself be treated as a meaningful level of causation?

This question has been debated in systems biology and related fields for years.

I do not think we need to choose between molecules and organization.

Molecular interactions generate organization.

But once organized, the system creates constraints on the behaviour of its components.

The architecture of a network influences which signals can propagate.

The structure of a cell influences which reactions can occur efficiently.

The spatial arrangement of molecules influences who can interact with whom.

The state of a network influences how it responds to subsequent signals.

In that sense, organization is not separate from molecular causation.

It is a higher-level description of the constraints created by molecular relationships.

The challenge is to turn this conceptual understanding into quantitative theory.

A theory must survive contact with data

I have become increasingly cautious about beautiful ideas.

Science is full of beautiful ideas.

Some are true.

Some are partly true.

Some are useful metaphors.

Some are simply wrong.

The difference is not determined by how attractive the idea sounds.

It is determined by evidence.

This is particularly important when we talk about information, quantum biology, computation, intelligence or consciousness.

These subjects are intellectually exciting.

That is precisely why they can become dangerous if speculation is allowed to outrun evidence.

The scientific approach should be:

What would this idea predict?

And then:

Can we test the prediction?

This is the principle I want to carry into any future theory of biological information.

What about quantum biology?

My own interest in quantum biology has reinforced this attitude.

I have explored the possibility that quantum phenomena may play deeper roles in biological systems than we currently understand.

Some quantum effects in biology are already well-established areas of research, while other proposed roles remain actively debated.

My own earlier work has also explored hypotheses about non-classical communication and possible quantum contributions to biological information processing. These ideas should be treated as hypotheses, not as established mechanisms. My paper on quantum phenomena in biological cells, for example, explicitly framed non-local or quantum-like biological communication as an alternative hypothesis arising from questions about intracellular organization and molecular crowding.

That distinction is extremely important to me.

There is a temptation to say:

“Everything is quantum, therefore biology is quantum.”

But that statement tells us very little.

A scientifically meaningful question is much narrower:

Is there a biological function for which quantum behaviour provides an explanation that cannot be adequately reproduced by a classical mechanism?

If the answer is yes, we need evidence.

If the answer is no, the quantum explanation should remain a hypothesis.

Recent discussions of quantum biology make essentially this methodological point: the field needs concrete biological functions, compelling evidence of nonclassical behaviour and experiments capable of distinguishing genuinely quantum mechanisms from classical alternatives.

I think this is an excellent model for the broader problem.

The same discipline should apply to information theory.

The danger of turning metaphors into theories

Words such as “information,” “computation,” “intelligence” and “memory” are extremely powerful.

But they can also become metaphors that sound scientific without actually being scientifically precise.

We can say:

“The cell computes.”

But what exactly does “compute” mean?

We can say:

“The genome contains information.”

But what kind of information?

We can say:

“The cell remembers.”

Where is the memory physically represented?

We can say:

“The network is intelligent.”

What measurable property qualifies as intelligence?

These are not objections to using such concepts.

They are invitations to define them more carefully.

If we want a new theory, we have to move from metaphor to measurement.

That means asking:

What is the variable?

How is it measured?

What is the mathematical relationship?

What are the boundary conditions?

What is the null model?

What alternative explanation exists?

What would falsify the hypothesis?

This is where biology can become more rigorous without becoming less imaginative.

What would a theory of biological information actually need?

This may be the central question of this article.

Suppose tomorrow someone told me:

“I have developed a new theory of biological information.”

My first question would not be:

“Is it beautiful?”

I would ask:

“What does it explain that we could not explain before?”

A useful theory should satisfy several requirements.

It should describe information not only as a quantity transmitted between components, but also as something transformed by a changing biological system.

It should incorporate **state**.

It should incorporate **context**.

It should incorporate **history**.

It should incorporate **time**.

It should incorporate **spatial organization** where relevant.

It should connect molecular interactions with network-level behaviour.

It should allow biological states to be represented mathematically.

It should generate measurable observables.

It should make predictions.

And those predictions should be testable.

Most importantly, it should coexist with molecular biology rather than pretending molecular biology is obsolete.

That is the kind of theory I would find worth pursuing.

A possible conceptual equation

At this stage, I find it useful to express the idea in a deliberately simple form:

$$O=f(I,S,H,C,T,L)$$

where:

- O represents an observable biological response,

- I represents incoming information or stimulus,
- S represents the current state of the system,
- H represents its history,
- C represents context,
- T represents time,
- L represents spatial location.

This is **not** being proposed as a finished equation for biology.

It is a conceptual scaffold.

It reminds us that the response of a biological system may depend on much more than the input alone.

A future theory would have to determine whether these variables can actually be defined and measured, whether some can be combined, whether additional variables are required, and whether the resulting framework provides predictive power.

Perhaps the final formulation will look completely different.

That would be perfectly acceptable.

The purpose of a conceptual equation is sometimes not to provide the final answer.

It is to make the question precise enough that we can start looking for an answer.

The missing variable may be organization

If I had to identify one idea that has increasingly occupied my thinking, it would be this:

Organization may itself be a biological variable.

We routinely measure concentration.

Expression.

Abundance.

Sequence.

Activity.

Mutation.

Connectivity.

But organization is often treated as something we see after the analysis rather than something we measure directly.

Suppose two systems contain the same components.

If their relationships differ, their behaviour may differ.

Suppose the same genes are expressed in two conditions.

If their coordination patterns differ, their system-level behaviour may differ.

Suppose the same network contains the same number of edges.

If the architecture of those edges differs, its functional properties may differ.

In each case, the parts are similar.

The organization is different.

Perhaps that difference deserves a quantitative description.

From “what is present?” to “how is it arranged?”

This might seem like a small shift in language.

It is actually profound.

Traditional molecular biology often asks:

What is present?

Systems biology asks:

How are the components interacting?

A theory of biological information might need to ask:

How is the system organized to transform information?

That is a different question again.

It brings together structure, dynamics, interaction and function.

It also suggests that the unit of analysis may sometimes need to change.

Instead of a gene.

Perhaps a module.

Instead of a protein.

Perhaps an interaction network.

Instead of an individual expression value.

Perhaps a trajectory.

Instead of a pathway.

Perhaps a dynamical state.

This does not make the smaller units irrelevant.

It simply adds another level of description.

Why this matters for disease

The implications could be particularly interesting in disease biology.

Disease is often described through molecular abnormalities.

A mutation.

A dysregulated gene.

An abnormal protein.

A metabolic defect.

A signalling pathway.

These descriptions are valuable.

But many diseases are also characterized by **system-level reorganization**.

Cancer, for example, is not simply a collection of mutated genes.

It involves changes in cellular state, signalling, metabolism, interactions with neighbouring cells, tissue organization and adaptation.

Similarly, stress responses are not simply a list of genes going up and down.

They involve coordinated changes across many biological layers.

This raises a possibility:

Could disease sometimes be understood as a transition between regions of biological state space?

Could early disease correspond to subtle changes in system organization before conventional markers become strongly abnormal?

Could system-level observables complement molecular biomarkers?

These are questions rather than conclusions.

But they are experimentally approachable.

The possibility of state-level biomarkers

This is one direction I find especially promising.

A conventional biomarker might be:

“Gene X is elevated.”

A state-level biomarker might instead say:

“The system has moved into a characteristic organizational state.”

That state could potentially be defined by several measurable properties.

For example:

- transcriptomic variability,
- network connectivity,

- spectral properties,
- information-sharing structure,
- temporal stability,
- coordination among biological modules,
- or other system-level observables.

The goal would not be to replace conventional biomarkers.

It would be to complement them.

Perhaps the future diagnostic system will say not only:

“This molecule is abnormal.”

but also:

“The organization of the biological system has shifted in a characteristic way.”

If such measures prove reproducible and predictive, they could become useful in disease biology.

But again, the important word is **if**.

Science has to earn such claims through validation.

Can we measure biological organization?

This is an area where I think biology has an opportunity.

We already have mathematical tools for measuring many aspects of complex systems.

We can quantify variability.

We can quantify entropy.

We can characterize network topology.

We can examine correlations.

We can study eigenvalues and eigenvectors.

We can analyze temporal patterns.

We can calculate information-theoretic quantities.

The challenge is not simply to calculate more numbers.

It is to identify **which numbers correspond to biologically meaningful properties**.

That requires comparison.

Different datasets.

Different species.

Different tissues.

Different stresses.

Different disease conditions.

Different experimental platforms.

And, importantly, different null models.

A proposed observable becomes scientifically interesting only when it demonstrates that it captures something robust and biologically interpretable.

Why “efficiency” and other ideas must wait for evidence

There are many tempting ways to describe biological organization.

We might talk about efficiency.

Robustness.

Coherence.

Complexity.

Order.

Resilience.

Information flow.

Adaptability.

But each of these words carries assumptions.

For example, if we say a biological system is “efficient,” we must ask:

Efficient at what?

Under what conditions?

At what energetic cost?

Compared with what alternative?

Likewise, if we say a system is “coherent,” we must define exactly what we mean.

This is why I would rather keep some of these ideas open for now.

Some concepts may eventually become useful quantitative constructs.

Others may turn out to be metaphors that help us think but cannot be rigorously measured.

Science should allow that uncertainty.

There is no need to reveal or name every emerging concept before it has been properly developed and tested.

Biology may live between order and disorder

This idea also changes the way I think about biological robustness.

We sometimes imagine robustness as perfect stability.

But a living system that never changes would not survive a changing environment.

At the same time, a system that changes completely in response to every disturbance would also be unable to maintain itself.

Life seems to require both:

stability and flexibility.

persistence and change.

memory and adaptation.

order and variability.

This tension may be one of the signatures of living systems.

A theory of biological information should perhaps therefore not ask simply:

“How ordered is the system?”

It should ask:

“How does the system organize variability while maintaining function?”

That is a much more interesting question.

The importance of time

Biology is also fundamentally temporal.

A gene may be expressed for minutes, hours or days.

A signalling pathway may activate within seconds.

A developmental programme may unfold over years.

Evolution acts over generations.

Different biological processes operate on different timescales.

The same molecular interaction can have different consequences depending on when it occurs.

A signal arriving before another signal may produce a different response from the same signal arriving afterward.

This means that temporal order itself can carry information.

A biological system is therefore not simply a spatial network.

It is a **spatiotemporal network**.

That is important because two systems with identical average measurements can have completely different dynamics.

The averages may be the same.

The trajectories may not be.

The importance of space

The same is true of spatial organization.

A molecule in one part of a cell is not necessarily equivalent to the same molecule in another location.

A signalling event at the membrane is not the same as the same event in the nucleus.

Two interacting proteins must be able to encounter one another.

A molecule's location can determine what information it can access and what processes it can influence.

This is one of the reasons cellular crowding and organization have been important themes throughout this series.

A cell is not a homogeneous reaction vessel.

It is spatially organized.

And if information depends on interactions, then spatial relationships can influence information processing.

So perhaps a biological information theory must eventually be both:

dynamic and spatial.

Information has a physical cost

There is another consideration that is easy to overlook.

Information processing is not free.

Biological systems require energy.

Maintaining gradients requires energy.

Building proteins requires energy.

Repairing damage requires energy.

Changing cellular organization requires energy.

Moving molecules requires energy.

Maintaining a stable state while responding to perturbations requires resources.

This means that biological information processing is embodied.

It occurs in physical matter.

It occurs within thermodynamic constraints.

It occurs in systems that must continually maintain themselves.

A future theory of biological information therefore cannot remain purely abstract.

It will ultimately have to connect information with physical implementation.

That does not mean every biological question must be reduced to thermodynamics.

It means that information in biology is not floating in an abstract mathematical space.

It is implemented by living matter.

And then there is evolution

Evolution adds another extraordinary layer.

A biological system is not designed by an engineer.

It is shaped by historical selection.

Its architecture is the result of accumulated changes.

Some changes persist.

Others disappear.

Some structures are repurposed.

Some constraints are inherited.

The result is an astonishing mixture of order, history and contingency.

This means that biological organization itself has an evolutionary history.

A network is not simply what it is today.

It is also the outcome of what happened before.

The same is true of genes, proteins, pathways, organisms and ecosystems.

This gives biological information a historical dimension that is difficult to capture in simple communication models.

The system carries information about its own past.

Evolution has written part of that history into the architecture of life.

Could evolution itself be viewed as information transformation?

This is another question I have found increasingly interesting.

Evolution changes biological information across generations.

But it does not simply copy information.

It generates variation.

It filters variation.

It interacts with environmental information.

It preserves some structures.

It eliminates others.

In this sense, evolution can be thought of as a long-term process of information transformation.

But again, we should be careful.

Saying “evolution is computation” can be a useful conceptual analogy, but analogy is not proof.

The important scientific question is whether computational or information-theoretic formulations generate useful predictions about evolutionary dynamics.

If they do, they become more than metaphors.

A theory should connect scales

One of the greatest challenges in biology is scale.

Molecules operate at one level.

Genes and proteins interact at another.

Networks create cellular states.

Cells interact to form tissues.

Tissues form organisms.

Organisms interact with environments and with one another.

Evolution operates over generations.

We have excellent theories and models at many of these levels.

The challenge is connecting them.

How does a molecular change become a network change?

How does a network change become a cellular state?

How does a cellular state become a phenotype?

How does a phenotype influence organismal fitness?

And how does selection feed back into molecular architecture across generations?

There is no reason to expect a single equation to answer all these questions.

But perhaps a broader information framework could provide a common language for describing transformations across levels.

That is what I mean when I talk about a new theory.

Not one equation replacing everything.

A framework connecting levels.

The cell as a transformation engine

I increasingly like a simple mental picture.

Imagine the cell not as a machine that receives instructions and executes them.

Imagine it as a **self-changing information transformation engine**.

It receives signals.

It generates signals.

It integrates information.

It modifies information.

It stores traces of past events.

It changes its own internal organization.

It responds to its environment.

And its responses alter the environment in return.

The cell is therefore both:

processor and product.

It processes information while continuously changing the machinery that performs the processing.

That is a remarkable property.

And perhaps it deserves a theoretical framework of its own.

The organism makes the problem even harder

Now move from the cell to the organism.

An organism contains enormous numbers of interacting cells.

Each cell has its own state.

Cells communicate.

Tissues communicate.

Hormonal signals travel.

Neural signals travel.

Immune signals travel.

Metabolic signals travel.

The organism is therefore a hierarchy of information-processing systems.

But it is not a simple hierarchy.

It is a network of networks.

A change at one level can propagate to another.

A systemic change can alter individual cells.

This is why reductionism and systems thinking should not be treated as enemies.

Reductionism tells us about the components.

Systems biology tells us about interactions and emergent organization.

A mature theory should help us move between the two.

This is not an argument against reductionism

I want to make this point particularly clearly.

There is sometimes a tendency to present molecular biology and systems biology as opposing philosophies.

I do not see them that way.

Molecular biology is indispensable.

Without molecules, there is no biological system.

Without chemistry, there is no molecular biology.

Without genes and proteins, there is no meaningful description of cellular mechanisms.

The problem is not reductionism itself.

The problem arises when we assume that explaining every component automatically explains the organization of the whole.

A bridge is made of steel.

Knowing the properties of steel is essential to understanding the bridge.

But knowing everything about steel does not tell you why the bridge has its particular shape.

Architecture matters.

Relationships matter.

Constraints matter.

Function emerges from organization.

Biology is not a bridge, of course.

But the analogy illustrates the conceptual point.

Perhaps biology needs a language of relationships

This may be one of the simplest ways to describe what I am searching for.

Molecular biology has developed extraordinarily rich languages for:

genes,
proteins,
sequences,
structures,
reactions,
pathways.

Systems biology has given us languages for:

networks,
modules,
interactions,
dynamics.

Information theory gives us languages for:

uncertainty,
entropy,
communication,
coding,
information.

Physics gives us languages for:

state,
energy,
dynamics,
phase transitions,
symmetry.

Perhaps the next theoretical step is not to invent an entirely new language.

Perhaps it is to find a way to connect these languages.

What if biological state is the bridge?

This is one possibility.

Imagine that we define a biological state using a combination of measurable variables.

The state might include molecular concentrations, network relationships, variability, spatial organization and temporal behaviour.

Then information would not simply be something transmitted between molecules.

It would be something that **changes the state of the system**.

The state would then influence how future information is processed.

This creates a dynamical framework:

Life is a process of state transformation.

And information is part of that transformation.

Could biological states have attractors?

Dynamical systems theory provides another intriguing possibility.

Many complex systems can have relatively stable regions in state space.

These are sometimes described as attractors.

Could biological states behave similarly?

A differentiated cell maintains a characteristic state.

A healthy physiological system maintains a range of states.

A disease process may push a system toward another region.

A perturbation may temporarily displace the system before it returns.

Development may involve transitions between relatively stable states.

If so, perhaps information-theoretic observables could help characterize the boundaries between these states.

What makes a system resilient?

What makes it fragile?

How much perturbation can it absorb before it crosses into another state?

Can early warning signals of such transitions be detected?

These are important questions in biology, medicine and agriculture.

They may also be approachable through public data.

This could change how we think about stress

Stress is a particularly interesting example.

Biologists often study stress responses by identifying genes that increase or decrease expression.

That tells us something important.

But stress is not a gene.

It is a **state transition**.

A system encounters a perturbation.

Its internal organization changes.

Some pathways become activated.

Others are suppressed.

Variability may change.

Interactions may change.

Metabolism may change.

The system attempts to adapt.

If successful, it returns toward a stable region.

If unsuccessful, damage accumulates or another state emerges.

This suggests a broader description:

Stress may be understood not merely as a collection of molecular responses, but as a trajectory through biological state space.

That is a very different way of looking at it.

And it is one that information theory may potentially help quantify.

The same idea applies to agriculture

As someone working in agricultural bioinformatics, I find this particularly relevant to crops.

Plants live in changing environments.

Drought.

Heat.

Salinity.

Pathogens.

Nutrient limitation.

Flooding.

Multiple stresses occurring together.

A plant does not respond to these conditions through a single gene.

It reorganizes.

Gene expression changes.

Metabolism changes.

Signalling changes.

Root architecture changes.

Hormonal networks change.

Development changes.

The phenotype that eventually appears is therefore the result of a system-level response.

This raises an agricultural question:

Can we detect the organizational state of a plant before visible stress symptoms appear?

Could transcriptomic organization provide an early signature?

Could network-level measures distinguish adaptation from damage?

Could a plant's trajectory through biological state space be used to predict resilience?

These questions could potentially connect fundamental information theory with practical agricultural bioinformatics.

The possibility of digital biological twins

This also points toward another future direction.

We increasingly hear about digital twins.

The idea is to create a computational representation of a physical system that changes as the physical system changes.

A biological digital twin would be enormously more complicated than an engineering one.

A living organism is not a machine with a fixed set of components.

It is adaptive, nonlinear, multiscale and historically contingent.

But perhaps information-theoretic state variables could help.

Instead of trying to model every molecule individually, a future digital biological model might represent the organism through a combination of:

molecular state,

network state,

metabolic state,

structural state,

environmental state,

and information-organizational state.

Such a representation would not capture everything.

It would be an abstraction.

But science works through useful abstractions.

The question is whether the abstraction predicts something that matters.

AI may help—but AI is not the answer by itself

Artificial intelligence provides powerful tools for dealing with complex biological datasets.

Machine learning can detect patterns humans may miss.

Deep learning can integrate enormous datasets.

Generative models can create biological sequences.

AI can assist in prediction, classification and simulation.

But AI also highlights the conceptual problem.

A model can predict accurately without necessarily telling us what biological information means.

Prediction is not automatically explanation.

A black-box model may tell us that a particular combination of features predicts disease.

But a theory of biological information should ideally tell us something about why those features matter and how they relate to the organization of the biological system.

Perhaps the future lies in combining the two.

AI for discovering patterns.

Theory for understanding those patterns.

Experiments for testing them.

This could create a powerful cycle:

Data → model → hypothesis → perturbation → new data → improved theory.

What would make a biological theory genuinely useful?

I would suggest a few practical criteria.

First, it should be **measurable**.

If we cannot measure the central concepts, we cannot properly test them.

Second, it should be **predictive**.

A theory should tell us what should happen under conditions that have not yet been observed.

Third, it should be **falsifiable**.

There should be observations that could prove it wrong.

Fourth, it should be **multiscale**.

It should help connect molecular, cellular and organismal descriptions.

Fifth, it should be **context-sensitive**.

It should recognize that biological responses depend on state and history.

Sixth, it should be **dynamic**.

Biology changes with time.

Seventh, it should be **perturbable**.

Its predictions should be testable through interventions, not only correlations.

And finally, it should be **compatible with established biology** while opening questions that existing frameworks do not answer comfortably.

That is a demanding list.

But perhaps that is exactly what a theory should be.

I am not looking for a grand theory of everything

There is a danger here.

Once we begin talking about information, quantum mechanics, intelligence and consciousness, it becomes tempting to search for one grand theory explaining all of biology.

I am not convinced that this is necessary.

Science often progresses by developing theories with clearly defined domains.

A useful biological information framework does not need to explain the universe.

It does not need to explain consciousness.

It does not need to replace evolution.

It does not need to replace molecular biology.

It simply needs to answer some questions better than our current descriptions do.

Perhaps the most useful theory is not the grandest one.

Perhaps it is the one that gives us a better question and a better experiment.

What I have learned from my own scientific journey

Looking back, the interesting thing is that I did not consciously set out to reach this point.

Molecular biology led me toward genes and proteins.

Bioinformatics showed me the scale of biological information.

Systems approaches showed me the importance of relationships.

Quantum biology raised questions about whether physical mechanisms deeper than classical descriptions might sometimes matter.

Information theory provided a language for thinking about uncertainty, communication and organization.

And gradually these ideas began to converge.

I began to see biology less as a collection of objects and more as a continuously changing information architecture.

I am not sure where this way of thinking will ultimately lead.

But I have become convinced that the questions are worth asking.

That may be one of the most important lessons of a scientific career.

You do not always end up with answers.

Sometimes you end up with better questions.

Science progresses when questions become measurable

There is a famous tendency in science to celebrate answers.

But scientific progress often begins much earlier.

It begins when someone asks a question that can be tested.

“Could this be true?”

“Could this variable be measured?”

“Could this mechanism be distinguished from an alternative?”

“Could this pattern predict what happens next?”

That is the stage at which I would place the ideas in this article.

I am not saying:

“This is how biology works.”

I am asking:

“Could biology work in a way that requires us to describe information, state and organization more explicitly?”

That distinction matters.

It is the difference between a hypothesis and a conclusion.

What would prove the idea wrong?

This is perhaps the most important question of all.

Suppose we develop a measure of transcriptomic organization.
We test it across dozens of datasets.
It performs well in the original dataset.
That is not enough.
We test it in another species.
It fails.
We test it under another stress.
It fails.
We control for batch effects.
The signal disappears.
We perform perturbation experiments.
Changing the proposed organization does not change biological function.
Then the hypothesis may be wrong.
That is not a failure of science.
That is science working.
A new theory should therefore come with its own possible failure conditions.
If it cannot fail, it cannot really be tested.

The role of negative results

This is particularly important in areas such as quantum biology.
Suppose we hypothesize that a biological phenomenon depends on a quantum mechanism.
A well-designed experiment may show that the phenomenon can be explained entirely through a classical mechanism.
That result is scientifically valuable.
It narrows the space of possibilities.
The same should happen with a biological information theory.
If a proposed organizational measure does not outperform simpler measures, we should abandon or modify it.
If a network property does not predict biological state, we should not force the interpretation.
If entropy adds nothing beyond ordinary variance, we should say so.
If a proposed state representation does not generalize, we should revise it.
The purpose of a theory is not to protect an idea.

It is to expose an idea to reality.

Perhaps the future lies in “state science”

We already have molecular biology.

We have structural biology.

We have systems biology.

We have synthetic biology.

We have computational biology.

Perhaps one day we will also have something that could broadly be called **biological state science**.

A science concerned not only with components, but with the measurable states and transitions of living systems.

Such a field might ask:

What defines a biological state?

How many dimensions are required?

Which dimensions are observable?

How stable is a state?

How does a system transition between states?

What information drives the transition?

What information is retained afterward?

How does organization change during the transition?

Which transitions are reversible?

Which are irreversible?

Which states correspond to health, stress, adaptation or disease?

Could these states be predicted?

These questions are not science fiction.

They are extensions of questions already being asked in systems biology, dynamical systems, computational biology and information theory.

What may be missing is a common framework that brings them together.

From state to trajectory

Perhaps even “state” is not enough.

A living organism is not simply a state.

It is a trajectory.

A seed becomes a plant.

A plant responds to drought.

A cell divides.

A tissue develops.

An organism ages.

A population evolves.

At every stage, the system carries traces of its past and possibilities for its future.

This makes biological time fundamentally different from a static database.

A genome sequence is relatively stable.

A living system is not.

The organism is a process.

Perhaps a future theory should therefore describe biological information as something that shapes **trajectories through state space**.

That gives us a much richer conceptual picture:

Information changes trajectories.

History influences trajectories.

Organization constrains trajectories.

Environment perturbs trajectories.

Biological function emerges from trajectories.

This may ultimately prove more useful than thinking only in terms of information transfer.

And perhaps “meaning” emerges from the trajectory

This brings us back to the question of meaning.

A nucleotide sequence can be represented as symbols.

A protein sequence can be represented as symbols.

A transcriptomic profile can be represented as numbers.

A network can be represented as a matrix.

But biological meaning is not necessarily contained in the representation alone.

Meaning emerges in relation to the system.

A sequence means something because of what it can do.

A signal means something because of how a biological system responds to it.

A mutation means something because of its consequences in a particular biological context.

A gene-expression change means something because of how it alters the state of the system.

This suggests that biological meaning may be fundamentally **relational**.

The information is not merely in the object.

It is in the relationship between:

signal, system, context and consequence.

That is a subtle but potentially powerful idea.

What if the unit of information is not the molecule?

This is perhaps one of the most provocative questions I would like to leave with the reader.

We often talk about information in biology as though it resides in DNA.

DNA certainly carries enormous amounts of biologically important information.

But perhaps DNA is only one layer.

Information can also reside in:

protein structure,

molecular concentrations,

network connectivity,

cellular architecture,

membrane organization,

epigenetic states,

spatial relationships,

temporal patterns,

and the history of a system.

Perhaps the fundamental unit of biological information is therefore not always a molecule.

Sometimes it may be a **relationship**.

Sometimes a **state**.

Sometimes a **pattern**.

Sometimes a **trajectory**.

This would be a substantial shift in perspective.

A theory of living systems may need to be relational

This is why I increasingly think that the future of biology may be less about discovering isolated objects and more about understanding relationships.

Genes relate to genes.

Proteins relate to proteins.

Cells relate to cells.

Cells relate to their environments.

Organisms relate to ecosystems.

Present states relate to past states.

Biological functions relate to organization.

And information relates to all of these.

Perhaps biology's deepest structure is therefore relational.

The parts matter.

But the relationships tell us what the parts can do together.

This also changes how we think about complexity

Complexity is often treated as a synonym for “many things.”

But a system can contain many components without being highly organized.

Conversely, a relatively small number of components can generate remarkably rich behaviour if their interactions are structured.

Therefore, complexity may not simply mean the number of components.

It may involve the structure of relationships.

This distinction matters enormously in biology.

A cell with thousands of proteins is not interesting merely because it contains thousands of proteins.

It is interesting because those proteins interact in organized ways that generate dynamic behaviour.

That is where information theory may have something important to contribute.

Information is not just quantity

Perhaps this is the deepest lesson I have taken from the series.

We can ask:

How much information is present?

But biology also forces us to ask:

Where is it?

When is it available?

Who receives it?

What state are they in?

What happened before?

How is it transformed?

What does it change?

What happens afterward?

These questions move us from information as a quantity toward information as a **process**.

And perhaps life itself is, at least in part, a process of organized information transformation.

I emphasize “perhaps.”

Because this is a hypothesis about how we might think about biology.

It is not a declaration about what life ultimately is.

Why I think this matters now

There is another reason I think these questions are timely.

Biology is entering an era of extraordinary data abundance.

Genomics.

Transcriptomics.

Proteomics.

Metabolomics.

Single-cell technologies.

Spatial biology.

Longitudinal measurements.

Imaging.

AI-generated biological models.

We are becoming exceptionally good at measuring biological systems.

But measurement creates a new problem.

How do we organize what we measure into understanding?

More data do not automatically produce better theory.

Sometimes they produce more correlations.

What we need are frameworks capable of turning large-scale measurements into meaningful descriptions of biological state and dynamics.

That is where I think information theory, systems biology, network science, dynamical systems and AI could intersect.

Not as competing disciplines.

As parts of a larger scientific language.

The next revolution may be conceptual

The great revolutions in biology have often involved new ways of seeing.

The microscope changed what we could see.

The cell theory changed how we understood organization.

Evolution changed how we understood biological diversity.

Genetics changed how we understood inheritance.

Molecular biology changed how we understood biological mechanisms.

Genomics changed the scale at which we could investigate life.

Now perhaps we are approaching another transition.

We can measure biological systems at enormous scale.

We can reconstruct networks.

We can model dynamics.

We can analyze information.

We can use AI to search spaces that are too large for humans.

The next revolution may therefore not simply be technological.

It may be **conceptual**.

We may need a better description of what all this information means when it is embedded in a living, changing system.

What I am proposing is deliberately unfinished

I want to end this article with an admission.

I do not know what the final theory will look like.

I do not even know whether there will be one unified theory.

Perhaps there will instead be several complementary frameworks.

Perhaps information theory will merge more deeply with systems biology.

Perhaps new mathematical concepts will emerge.

Perhaps biological state will become a major unit of analysis.

Perhaps network organization will provide the bridge.

Perhaps none of these ideas will survive in their present form.

That is completely acceptable.

A research programme should be allowed to evolve.

The purpose of a hypothesis is not to be permanently correct.

It is to guide us toward experiments.

What I would like to see next

I would like to see biological datasets analyzed not only for individual molecular changes, but for changes in **organization**.

I would like to see biological states represented quantitatively.

I would like to see trajectories rather than only snapshots.

I would like to see perturbation experiments designed to test whether organizational properties have functional consequences.

I would like to see information-theoretic measures compared rigorously with existing network and transcriptomic measures.

I would like to see such approaches tested across species and datasets.

I would like to see negative results reported honestly.

I would like to see AI used not merely to predict biological outcomes but to discover candidate theoretical variables.

And I would like to see quantum biology treated with the same intellectual discipline:

interesting enough to investigate, but never too interesting to test.

From molecular biology to a biology of relationships

When I look back at the path that brought me here, I see a gradual change.

I began with molecules.

Then came genes.

Then networks.

Then large-scale biological data.

Then information.

Then questions about computation.

Then questions about intelligence.

Then questions about consciousness.

But perhaps the deepest transition is simpler than all of these.

It is the transition:

from things to relationships.

A gene is important.

But its relationships matter.

A protein is important.

But its interactions matter.

A cell is important.

But its organization matters.

An organism is important.

But its relationships with other organisms and its environment matter.

Information is important.

But how information is transformed by the system may matter even more.

The question I am left with

After all these years of studying biology from different perspectives, I keep returning to one question:

What exactly makes a biological system biological?

Is it its molecules?

Certainly, in part.

Is it its chemistry?

Absolutely.

Is it its genetic information?

Yes.

But perhaps there is something else.

Perhaps it is the way those components are organized.

The way they communicate.

The way they remember.

The way they respond.

The way they adapt.

The way they maintain themselves while constantly changing.

The way information becomes action.

The way action changes the system.

And the way the changed system encounters the next piece of information.

That is a loop.

And perhaps life is, at least in part, the continuous operation of that loop.

The beginning of a different kind of biology

I began this series by asking whether biology needs a new information theory.

Ten articles later, I think the question has become more precise.

Perhaps what biology needs is not simply a theory of **information**.

Perhaps it needs a theory of **information transformation in self-organizing, self-changing systems**.

Such a theory would not replace molecular biology.

It would build on it.

It would not reject chemistry.

It would give chemistry another level of interpretation.

It would not replace genetics.

It would ask how genetic information interacts with state, context, history and organization.

It would not replace systems biology.

It would try to provide a more quantitative language for some of the questions systems biology has already raised.

And it would not turn every biological mystery into quantum mechanics.

It would ask, carefully and experimentally, where quantum effects genuinely matter.

That is the kind of theory I would like to see.

And perhaps, in some small way, the kind of theory I would like to spend the coming years helping to explore.

One final thought

Science often advances by discovering new objects.

A new gene.

A new protein.

A new molecule.

A new pathway.

A new species.

But sometimes the bigger discoveries come from realizing that we have been asking the wrong kind of question.

Perhaps the future of biology will not be defined only by discovering **more things**.

Perhaps it will be defined by discovering **better variables**.

Variables that tell us how a living system organizes itself.

Variables that tell us how information changes the system.

Variables that tell us how the system changes the meaning of subsequent information.

Variables that allow us to describe transitions between biological states.

Variables that connect molecules to networks, networks to cells, cells to organisms, and organisms to environments.

And perhaps, one day, these variables will allow us to describe something that has always been difficult to capture:

not just what life is made of, but how life remains organized while constantly changing.

That, for me, is the real reason for **Rethinking Biology**.

Not because the biology we have is wrong.

But because it has become rich enough to make us ask questions that our older descriptions were never designed to answer.

And perhaps the next revolution in biology will not begin with the discovery of another molecule.

Perhaps it will begin with the discovery of a new way to describe **how living systems organize, transform and use information**.

The theory is not finished.

Perhaps it has not even been fully formulated yet.

But the question is now clear.

And sometimes, in science, that is where the most interesting journeys begin.

Scientific note

The ideas in this article deliberately occupy different levels of certainty. Molecular biology, information theory, systems biology and network analysis provide established scientific foundations. The suggestion that biological **state, organization, information transformation and system-level observables** could form the basis of a broader theory is a proposed research direction, not an established theory. Likewise, the possible functional importance of quantum effects in biological

information processing remains a question requiring discriminating experimental evidence. This distinction between established knowledge, active scientific questions and exploratory hypotheses is essential to the spirit of *Rethinking Biology*.

The series plan deliberately places Article 10 as the author's personal intellectual journey—from **molecular biology → bioinformatics → quantum biology → information theory**—after the earlier articles have established the scientific background.

The End of a Series — But Not the End of the Question

Ten articles ago, I began this journey with a simple question:

Does biology need a new way of thinking about information?

I did not expect that question to take me from molecules and genes to computation, intelligence, collective behaviour, consciousness and, finally, to the possibility of a new theoretical framework for biological information.

I have not arrived at a final theory.

Perhaps that is appropriate.

Science rarely ends with the final answer. More often, one journey ends by opening another.

For me, *Rethinking Biology* has reached that point.

The series has travelled from:

Molecules → Information → Computation → Intelligence → Collective Intelligence → Consciousness → A New Way of Thinking About Biological Information

Along the way, I have explored questions about cells, biological information, quantum biology, computation, intelligence, collective behaviour and consciousness.

And Article 10 has brought me back to the question that lies beneath all of them:

How do living systems organize, transform and use information while continuously changing themselves?

I have deliberately not presented a finished theory.

Instead, I have tried to identify questions that may be worth pursuing:

How should biological state be described?

Can biological organization be quantified?

Can we move from studying individual components to studying relationships and system-level states?

Can information theory help us understand biological dynamics?

Can we develop new observables that complement conventional molecular measurements?

Can such ideas generate predictions that can actually be tested?

These questions will continue to interest me.

But I think they belong to the next stage of the journey.

So, with **Article 10**, I am bringing the *Rethinking Biology* series to a close.

Not because the questions have been answered.

Quite the opposite.

Because I think we have reached the point where the questions themselves have become more interesting than the answers.

The series may end here, but the exploration does not.

Perhaps answering these questions will require a biology that is not less molecular, but more connected—a biology that can move naturally from **molecules to relationships, from relationships to organization, and from organization to biological function**.

This is where I leave the series.

Not with a theory, but with a question.

And perhaps that is the best place for a scientific journey to end.